

Electrical Network Analysis (EE2004)

Date: September 21st, 2024

Course Instructor(s)

Mr. Abdul Majid

Ms. Beenish Fatima

Mr. Haris Mujtaba

Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 40

Total Questions: 2

Roll No

Section

Student Signature

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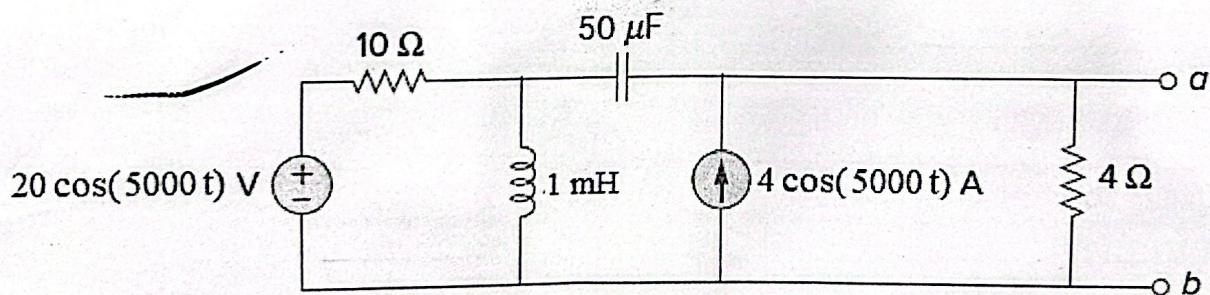
1. Attempt all the questions.
2. Attempt all parts of the same question together.
3. Show all the steps with proper labelled circuit diagrams, and answers with proper units.

CLO #1: Apply phasor-domain analysis to solve circuits containing R, L, C, and mutual inductance

Q1: Solve the circuit below to

[20 marks]

- a) Construct the frequency domain circuit
- b) Construct the Thévenin Equivalent circuit w.r.t. terminals 'a' and 'b' using source transformations
- c) Determine the steady state expression of Thévenin Equivalent voltage



CLO #1: Apply phasor-domain analysis to solve circuits containing R, L, C, and mutual inductance

[20 marks]

Q2:

- Calculate the current I_o in the circuit shown below using node voltage method
- Determine the expression of $i_o(t)$

